

CSE 4616 – Wireless Networks

Lab 03: In-Lab Assessment Tasks

Duration: 1 hour 30 minutes — Individual Assessment

Tools: OMNeT++ and INET Framework

General Instructions

- All tasks build on the **WirelessMediumStudy** network and the `WirelessMediumBaseline` INI configuration from the lab manual.
- Unless stated otherwise, start from the baseline and apply only the modifications specified in each task.
- Name your files: `TaskX_StudentID.ned` and `TaskX_StudentID.ini`.
- Take Qtenv screenshots *while the simulation is running* (between $t = 50$ s and $t = 150$ s) unless stated otherwise.
- Write all analysis answers in your lab report.

Baseline Configuration Reminder:

- 4 wireless hosts + 1 access point, all on same subnet
- 2 apps per host: `app[0] = UdpEchoApp` (port 5000), `app[1] = UdpBasicApp`
- `messageLength = 1000B`, `sendInterval = 0.1s`, `stopTime = 180s`
- `cwMin = 15`, `cwMax = 1023`, `rtsThreshold = 3000B` (RTS/CTS off), `retryLimit = 7`
- `txPower = 20mW`, `bitrate = 54Mbps`, `frequency = 2.4GHz`

Task 01: Observe the Effect of Contention Window Size

Background: The contention window (cwMin) controls how spread out backoff choices are. A small cwMin means devices pick similar backoff values, increasing collision probability. A large cwMin reduces collisions but adds unnecessary delay.

Required Modifications (INI file only):

Create **three separate INI configurations** by adding three [Config] sections:

```

1 [Config CWmin_Small]
2 extends = WirelessMediumBaseline
3 **wlan[0].mac.dcf.channelAccess.cwMin = 3      # Very small window
4
5 [Config CWmin_Default]
6 extends = WirelessMediumBaseline
7 **wlan[0].mac.dcf.channelAccess.cwMin = 15     # 802.11g default
8
9 [Config CWmin_Large]
10 extends = WirelessMediumBaseline
11 **wlan[0].mac.dcf.channelAccess.cwMin = 63    # Large window

```

Listing 1: Three CWmin configurations

Run each configuration separately and record your observations.

Observation Table (fill in during/after each run):

Metric	cwMin = 3	cwMin = 15	cwMin = 63
Packet animations in Qtenv (sparse/ dense?)			
Any noticeable pauses / backoff delays?			
Console: any re- transmission mes- sages?			
Qualitative throughput (high- /med/low)			

Table 1: Observation table for Task 01

Analysis Questions:

- Why does a smaller cwMin increase the probability of collisions when four hosts compete for the channel simultaneously? Use the backoff selection mechanism in your explanation.

- b. With `cwMin = 3`, how many distinct backoff values can a device choose from on its first transmission attempt? Calculate this.
- c. Based on your observations, which `cwMin` value gave the best apparent performance for this four-host scenario? Justify your answer.
- d. The 802.11 standard fixes `cwMin = 15` for 802.11g. Why might a larger value not always be better, even in a dense network?

Deliverables

- **Modified INI file** with all three configurations: `Task1_StudentID.ini`
- **Screenshot** from each of the three Qtenv runs (3 screenshots total), labelled with the configuration name
- **Completed observation table** and **written answers** to questions (a)–(d)

Task 02: Enable RTS/CTS and Observe Its Effect

Background: In the baseline, RTS/CTS is disabled because the threshold (3000B) exceeds the packet size (1000B). In this task you will enable RTS/CTS by lowering the threshold below the packet size, and compare behaviour with and without it.

Required Modifications (INI file only):

Create **two** configurations:

```
1 [Config RTS_Off]
2 extends = WirelessMediumBaseline
3 # rtsThreshold is 3000B in baseline, larger than 1000B packet = RTS
  /CTS off
4
5 [Config RTS_On]
6 extends = WirelessMediumBaseline
7 # Lower threshold below packet size to force RTS/CTS on all data
  frames
8 **wlan[0].mac.rtsThreshold = 100B
```

Listing 2: RTS/CTS off vs. on

Additionally, for RTS_On, reduce the transmission power to **simulate a hidden terminal scenario** by placing hosts at the edge of each other's range:

```
1 # In RTS_On config: reduce power so not all hosts hear each other
2 **wlan[0].radio.transmitter.power = 1mW
```

Analysis Questions:

- Describe the full RTS/CTS exchange sequence. How many frames are exchanged before a single data frame is successfully delivered? Draw a timeline: Sender → AP → Receiver.
- In the RTS_On run, did you observe any difference in packet delivery compared to RTS_Off? Describe what you saw in Qtenv.
- RTS/CTS adds overhead (extra frames). Under what network conditions is this overhead *worth paying*? When is it *not worth it*?

Deliverables

- **Modified INI file:** Task2_StudentID.ini
- **Two screenshots:** one from each configuration, taken at similar simulation times for comparison
- **Timeline diagram** for question (a) drawn by hand or digitally
- **Written answers** to questions (b), (c), and (d)

Task 03: Compare 802.11b vs. 802.11g Parameters

Background: Different 802.11 amendments use different PHY and MAC timing parameters. In this task you will configure one simulation to use **802.11b** parameters and another to use **802.11g** parameters, and compare their behaviour.

Required Modifications (INI file only):

```
1 [Config Wifi_80211b]
2 extends = WirelessMediumBaseline
3 # 802.11b PHY
4 **.wlan[0].radio.transmitter.bitrate      = 11Mbps
5 **.wlan[0].radio.carrierFrequency         = 2.4GHz
6 # 802.11b MAC timing
7 **.wlan[0].mac.dcf.channelAccess.cwMin    = 31
8 **.wlan[0].mac.dcf.channelAccess.slotTime = 20us
9
10 [Config Wifi_80211g]
11 extends = WirelessMediumBaseline
12 # 802.11g PHY (same as baseline, shown explicitly for comparison)
13 **.wlan[0].radio.transmitter.bitrate      = 54Mbps
14 **.wlan[0].radio.carrierFrequency         = 2.4GHz
15 # 802.11g MAC timing
16 **.wlan[0].mac.dcf.channelAccess.cwMin    = 15
17 **.wlan[0].mac.dcf.channelAccess.slotTime = 9us
```

Listing 3: 802.11b vs. 802.11g parameter sets

Calculation Exercise:

Using the timing formulas from the lab manual, calculate the following for **both** configurations. Show all working.

Analysis Questions:

- Based on your calculations, how many times faster is a single 802.11g frame delivered compared to 802.11b for the same 1000B packet?
- In Qtenv, did the **visual packet animation rate** differ between the two configurations? Describe what you observed.
- 802.11b has a larger `cwMin` (31 vs. 15) despite being older and slower. Why might a larger contention window have been chosen for the slower standard?
- If you were designing a wireless network for an environment where devices are mostly stationary and close together, which amendment's parameters would you prefer? Justify your answer.

Deliverables

- **Modified INI file:** Task3_StudentID.ini
- **Two screenshots:** one per configuration at comparable times

Metric	802.11b	802.11g
Frame duration for 1000B at given bitrate		
DIFS duration ($= \text{SIFS} + 2 \times \text{SlotTime}$)		
Average backoff ($\text{CW}_{\text{min}} / 2 \times \text{SlotTime}$)		
Total channel time per frame (approx.)		
Max theoretical frames/s ($1 / \text{total time}$)		

Table 2: Timing calculation table for Task 03

- **Completed calculation table**
- **Written answers** to questions (a)–(d)

Task 04: High-Load Contention and Binary Exponential Backoff

Background: Binary Exponential Backoff (BEB) is the collision recovery mechanism in CSMA/CA. Each time a collision occurs (no ACK received), the contention window doubles. In this task you will create an extreme high-load scenario to force many collisions and observe BEB in action, then compare different `retryLimit` and `cwMax` settings.

Required Modifications:

NED file: Add a **sixth host** (host5 and host6) to the `WirelessMediumStudy` network to increase contention:

```

1 host5: WirelessHost {
2     parameters:
3         @display("p=200,200");
4 }
5 host6: WirelessHost {
6     parameters:
7         @display("p=300,200");
8 }

```

Listing 4: Add host5 and host6 to `WirelessMediumStudy.ned`

INI file: Create two configurations with very high load and different BEB limits:

```

1 [Config BEB_Aggressive]
2 network = WirelessMediumStudy
3 sim-time-limit = 200s
4 **.addDefaultRoutes = false
5
6 **.*host*.numApps = 2
7 **.*host*.app[0].typename = "UdpEchoApp"
8 **.*host*.app[0].localPort = 5000
9 **.*host*.app[*].typename = "UdpBasicApp"
10
11 # All 6 hosts send to each other (round-robin)
12 **.host1.app[1].destAddresses = "host4"
13 **.host2.app[1].destAddresses = "host5"
14 **.host3.app[1].destAddresses = "host6"
15 **.host4.app[1].destAddresses = "host1"
16 **.host5.app[1].destAddresses = "host2"
17 **.host6.app[1].destAddresses = "host3"
18
19 **.*host*.app[1].destPort = 5000
20 **.*host*.app[1].messageLength = 1000B
21 **.*host*.app[1].sendInterval = 0.02s # Very high load: 50 pkt/s
    per host
22 **.*host*.app[1].stopTime = 180s
23
24 **.wlan[0].radio.transmitter.power = 20mW
25 **.wlan[0].radio.transmitter.bitrate = 54Mbps
26 **.wlan[0].mac.dcf.channelAccess.cwMin = 15

```

```
27 **wlan[0].mac.dcf.channelAccess.cwMax      = 63      # Small max =  
    aggressive retries  
28 **wlan[0].mac.dcf.channelAccess.retryLimit = 3      # Drop quickly  
29  
30 [Config BEB_Patient]  
31 extends = BEB_Aggressive  
32 **wlan[0].mac.dcf.channelAccess.cwMax      = 1023   # Large max =  
    patient backoff  
33 **wlan[0].mac.dcf.channelAccess.retryLimit = 7      # More retries
```

Listing 5: High-load BEB configurations

Analysis Questions:

- In BEB_Aggressive, after 3 collisions on the same frame, what is the contention window size? Show the doubling sequence: CW after attempt 1, 2, 3.
- In BEB_Patient, after 7 failed attempts, what happens to the frame? What does this mean for the application sending it?
- In Qtenv, compare the two runs. In which configuration do you see more continuous packet flow? Which has more visible pauses? Explain why in terms of BEB behaviour.
- A small cwMax means the window stops growing quickly after collisions. What is the **risk** of this in a very congested network? What is the **benefit**?
- Based on what you observed, propose a set of cwMin, cwMax, and retryLimit values that you believe would give the best balance of low delay and low packet loss for this 6-host scenario. Justify your choice.

Deliverables

- **Modified NED file:** Task4_StudentID.ned (with host5 and host6 added)
- **Modified INI file:** Task4_StudentID.ini
- **Two screenshots:** one per configuration showing the high-load simulation running
- **Written answers** to questions (a)–(e)